

Sample University Syllabus - Data Structures & Algorithms

Course: Data Structures & Algorithms

Semester: Fall 2026

Credits: 4

Course Outcomes

Students will understand fundamental data structures, algorithmic techniques, time complexity analysis, and problem-solving strategies.

Unit 1: Complexity Analysis

Weightage: 15% **Assessment Date:** 2026-09-10

- Introduction to Algorithms
- Asymptotic Analysis (Big-O, Big-Theta, Big-Omega)
- Recursion and Recurrence Relations

Unit 2: Linear Data Structures

Weightage: 20% **Assessment Date:** 2026-09-24

- Arrays
- Linked Lists
- Stacks
- Queues

Unit 3: Trees

Weightage: 25% **Assessment Date:** 2026-10-15

- Binary Trees
- Binary Search Trees
- Heap
- Trie

Unit 4: Graphs

Weightage: 20% **Assessment Date:** 2026-11-05

- Graph Representation
- BFS
- DFS
- Shortest Paths

Unit 5: Advanced Topics

Weightage: 20% **Assessment Date:** 2026-11-26

- Greedy Algorithms
- Dynamic Programming
- Backtracking

Assessment Pattern

Assignments: 20%

Mid Semester Exam: 30%

End Semester Exam: 50%